GROUP WORK PROJECT # _3__
GROUP NUMBER: _10177___________

MScFE 660: RISK MANAGEMENT

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Project 3

Step 1

Defining the datasets.

1.a. Student A writes 1–2 pages defining the purpose of the training set.

Before we run and fit the mode on our cleaned and refined dataset which has no errors or inconsistencies like missing data. We split the dataset onto three components: training, validation and testing set.
The training set serves as a foundational back bone and this is the first step to train the model hence the name train data. The model will learn from the train data with defined hyper parameters. It will found the patterns, relationships and predictions. This may result in output like no of trees (in random forest), coefficients in regression model, or conditional probabilities or edges and nodes in case of graphical model.
If the model performance as accuracy score is low for training dataset then we need to look into this issue by changing hyper parameters or including more meaningful variables or removing redundant variables. In the context of forecasting crude oil prices using probabilistic graphical model (PGMs) the training set plays a critical role.

The main purpose of the training set is to first establish the structure of model catering to the oil market like macro economic indicators, physical supply like production, oil prices and geo political factors, based on train data, on which it will test its unseen validation and test data. The validation and test data will check whether the model has not been over crammed from the data. So that it underperforms on new and unseen validation and test data. We may need to look at improving generalization or issues like bias variance tradeoffs. Once we train the model we will not change the underlying equations and the model will be tested on what it has learned from the train data.

2.

The training dataset allows the bayesian belief network or hidden markov models to understand the causal dependencies or conditional probabilities among variables that affect crude oil prices. This is the knowledge it has learned by running the model on train data.

This can be used to

- Estimate the parameter on conditional probability distribution (parameter learning)
- Estimate the structure of Graphical model (structure learning)
- Serves as the basis for discretizing time series data and identifying market regimes (bullish, bearish)..
- Help tune the model on techniqueslike maximum likelihood estimation or Bayesian Estimation

The whole dataset as per the research paper set is pulled from FRED and EIA. We can get individual series data like GDP, CPI, industrial production, interest rates etc. along with Oil prices. After removing inconsistencies in the series we can join them to form a dataset. Form which we will get the train,validation and test dataset.

The individual series are pulled via API and are monthly time series. They can be used for Discrete Bayesian belief network or hidden markov models.

- Learning relationships: This can help determine how various factors like oil supply, interest rates and inflation affect each other and affect oil prices
- Model Calibration: before the model is tested and validated. It needs to be calibrated. Repeatability and consistency of results is also important.
- Reducing Human Bias:Rather than relying on expert opinion or intuition, this allows learning form data driven approach, minimizing inconsistencies due to subjective analysis
- Preparing the inference: The better and accurate learning from data, the better will be forecast or predictions

The learning form training set is a structured representation of how crude oil prices behaves based on historical patterns and economic theory and how they are affected by other factors. The design and content of the training set impacts the reliability and findings and interpretability of bayesian frameworks underlying.

2.

1.b. Student B writes 1–2 pages defining the purpose of the validation set.

Sometimes the model underperforms on the test dataset. One of the reasons may be not the required hyper parameter or the model did not converge to the required minimum loss optimization function properly. Se we may need to tweak the model hyper parameters. So we do it in steps, we train the model on train data and then test the validation test. Once we are satisfied with the performance we then test the model on test dataset. The validation set step is in between training the model and test the model on test set. The validation step can play an important role in model tuning, performance evaluation and generalization to avoid over fitting .

The validation set is used to tune the model parameters and rerun the model for learning new structure, it is also used to evaluate the model performance on one part of unseen data. This helps in assessing whether the relationships and probability inference are likely to perform and generalize to further future real world data. Both training or fitting the model and validating the model to check performance on validation data are part of the training phase.

In this research, the validation set helps answer questions like:

- Is the Bayesian Network overfitting to noise in the training phase or cramming even the finer details like rapid volatile swings or certain short term random deviations .
- Does the model capture macroeconomic shifts and energy supply/demand dynamics in a way that are predictive outside the training window?
- Whether the model hyperparameters (e.g., discretization bins, regularization settings) are optimal or not?

Purpose

1. **Model Selection and Hyperparameter Tuning**
   We may implement multiple models and finalize the final one after validation testing. We can try different scoring functions for structure learning, varying no of hidden states or using alternative discretization techniques. The validation test results will help us to compare the models and choose objectively or subjectively. This will bring us close to have the model which can generalize and perform in unseen data of real world, or future data and also maintain a good bias variance balance

2. **Preventing Overfitting**
   Overfitting occurs when the model performs well on a train dataset but not on unseen data which is next in time series. The validation allows us to detect it and find the cause. It may be possible that the model as learned spurious correlations or the way binning of variable is done is not appropriate and fittign the purpose and we may need to check on it.

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It may be possible that the validation and test dataset has data pertaining to COVID-19 or 2007-08 stock market crash. This may not be predicted in advance. SInce the train ing phase learning has dependency on the historical data present in the train set. Another example could be stock splitting or the company buying its own stocks and it will affect its market running performance.

3. **Testing Temporal Stability**
   The predictions and performance of the model should be consistent for all the train, validation and test sets. The model should adapt to change in regimes like bull to bear or bear to bull due to the impact of change in OPEC policy or global economic crisis. For e.g. data from 2000-2010 might be used for training while data form 2011-2014 could be use for validation set.
4. **Improving Inference Quality**
   The validation set provides us how well the Bayesian Network or HMM is making probabilistic inferences. For e.g., if the model consistently misunderstood the importance of volatility or misjudges the direction of price changes on the validation set, it may indicate poor prior assumptions or faulty conditional probability tables.

According to the dissertation, the validation set is likely constructed **after data cleaning and discretization** but **before final testing**. The time-series is sourced from sources like the EIA and FRED. As such, a time range aware (index based) split is used, meaning that past historical data (training) precedes further time series data with no overlapping (validation) and it precede the future test dataset, preserving temporal causality. The **validation error** such as trend deviation from actual oil prices or failure to pick pace with unpredictable volatility or prediction confidence guides us to whether to retrain the model with a new tweaked train dataset or use a new model or adjust hyper parameters.

The **validation set functions as a diagnostic tool** that ensures the probabilistic model is robust, stable, credible and ready to go ahead for final testing and real world deployment. It enables model optimization without checking on the final testing process. In the volatile and complex domain of oil price forecasting, where new macroeconomic and geo-political conditions emerge, the validation set helps ensure that the model is not just historically accurate but can cope with changing conditions and is also **forward looking and reliable**.

Leave-One-Out-Cross-Validation (LOOCV):

- In the **Validation** section (Section 4.3), the paper repeatedly removes **one observation** from the dataset, trains the Bayesian Network on the remaining data, and then predicts the left-out value.

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- The observation is then replaced back (with replacement) to the training set and then tested for next one. This process is done for each data point in turn, i.e. each sample is used only once as the validation set and n–1 times as part of the training se**t**.
- The performance metrics (MAPE, directional accuracy) are then averaged over all iterations.

1.c. Student C writes 1–2 pages defining the purpose of the testing set

It is important that the model trained not only performs well on validation data but also on unseen test data which was not part of learning i.e. training set. In the context of crude oil price predictions using Probabilistic Graphical Models, Bayesian Networks and Hidden Markov models. Once we have passed the testing phase we can use our predictions for our next step for our application like presenting graphs or using them for further management decisions or decisions like investment planning

Distinction from Training and Validation Sets
While the training set helps to learn and fit the model on data and the validation helps to fine tune the model such as, choosing the number of states in HMM or no of discrete bins for Discrete Bayesian Network etc. The testing phase uses a separate test dataset which was not part of training or validation step. It should be tested only once after the model performance has successfully passed the validation testing. It is used to evaluate how well the model performs on unseen new data. It can be used for future data predictions.

In the context of time series forecasting and this paper, the test set represents a future time window that the model must predict based on the patterns, equations and insights learned by the model during training.

Functions of the Testing Set
Final Evaluation
The testing set provides an isolated and measure of generalization vs model performance. Metrics such as prediction accuracy, error rate, or confidence interval coverage are computed using this set.

Assessing Generalization Ability
In machine learning we should build models that performs well and also generalizes well to new situations.. If a model performs well on the training and validation sets but poorly on the testing set, this represents overfitting, i.e. the model or the dataset is still missing something and is incomplete and has failed to capture the general patterns and has memorized noise or less important details in the training data.

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Backtesting
In financial modeling, the testing set is also referred to as backtesting data. It is used to simulate how the model would have performed if it had been deployed during a specific time period. For e.g., testing the model on data from 2015–2018 after training it on 2000–2014 allows us to observe whether it could have anticipated or handled major events like the 2016 oil price crash or the OPEC production cuts.

Stress Testing and Scenario Analysis
The paper also discusses stress testing e.g analyzing how the model performs under extreme or unusual economic conditions. The testing set can include data from time periods of such events (e.g., financial crises, war in middle east, imposed import tariffs. geopolitical shocks) to observe whether the model maintains predictive stability or breaks down under pressure. This is vital for decision-makers who wants to rely on model outputs during such periods.

Benchmarking Against Traditional Models
The research paper compares PGMs to more conventional models like GARCH, which assume certain statistical properties such as constant volatility structures. The testing set provides a ground for performance comparison across these different modeling frameworks. By evaluating prediction errors or forecast consistency on the same testing set, one can determine whether the PGM approach provides a meaningful solution or shows an improvement.

It may also be possible that predictions of one model is performing well in certain time period of testing set while another model performs in another part of time period in testing set. The reason and solution to the issue should be explored.

Construction and Use
The testing set is typically chosen from the most recent segment of the dataset, following the training and validation periods in a strict chronological order. For example:

Training set: 2000–2010

Validation set: 2011–2014

Testing set: 2015–2018

This reflects the requirement for testing set that we can only make predictions about the future using learning from past historical data.

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The testing set should represent the real world data in which a model is expected to perform. It validates whether the entire effort in the modeling process : data collection, cleaning, fitting the model on train data, structure learning, parameter estimation, and validation was successful. In the volatile and globally influential crude oil market, a model that performs well on a properly constructed testing set can be trusted to assist in making the right investment, trading, or risk management or policy decisions. So, the testing set provides us the **final most trustworthy evidence** of a model's effectiveness.

Step 2

Comparative Analysis: Validation vs. Testing Sets

Based on the paper "Application of Probabilistic Graphical Models in Forecasting Crude Oil Price"

With respect to section 4.3 of the paper, we compare and explore how these datasets differ in purpose, and role within the PGM modeling workflow. We will discuss on the allocation strategy used to create these sets.

1. Dataset Allocation Strategy

The research paper proposes a time-series split that aligns with non overlapping periods and chronological integrity, ensuring that data used to forecast future events is based only on past information, a practice for forecasting models.

The **overall dataset** consists of macroeconomic and market specific time series indicators, sourced from:

- **FRED** (Federal Reserve Economic Data)
- **EIA** (Energy Information Administration)

All series are brought to be sampled at a **monthly frequency**.Once the series are brought at same frequency. They are joined and cleaned like removing '-' or blanks with nans and filling nans using backward filling. Having Date as index of dataframe

Dataset	Time Range	Purpose	Usage
Training Set	2000 – 2010	Model learning: parameter estimation & structure	Learning

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Validation Set	2011 – 2014	Hyperparameter tuning, model selection	Optimization
Testing Set	2015 – 2018	Final model evaluation on unseen data	Performance, Generalization,

This partitioning or splitting of the original dataset ensures that future data is never leaked into the model learning process, and learning data is not leaked into validation and testing set preserving the credibility of performance evaluation.

2. Purpose & Role

2.1 Validation Set

The validation step is used after the model fitting on train data step to:

- Evaluate various configurations (e.g., different structure learning algorithms,different scoring functions used as Bdeu, K2 or Aic, this will help determine the best network structure in the context of Bayesian Networks)
- Tune hyperparameters (e.g., number of discretization bins).
- Prevent overfitting by exposing model weaknesses before final testing.
- Select between alternative models based on performance metrics like Mean Absolute Percentage Error (MAPE) or directional accuracy.

It acts as an **internal checkpoint**, helping shape the final version of the model.

2.2 Testing Set

The testing set is used after the model has been finalized and successfully passed the validation testing. This is one of the most important step before we conclude our analysis and model implementation, It serves to:

- **Simulate deployment** by testing performance on unseen, forward looking future data.
- Evaluate **generalization capability** vs model performance even in complex situations in real world and changing environment.
- Quantify out-of-sample accuracy using the same performance metrics.

This is the **final audit** of the model, used to decide whether it is fit for operational use in financial forecasting.

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3. Comparison: Validation vs. Testing Sets

Feature	Validation Set	Testing Set
Used during	Second step of Model training phase	After final model is selected and passed the validation phase testing
Purpose	Tune hyperparameters and select model	Evaluate generalization and performance
Accessible during training?	Yes, but only for optimization, not learning	No (completely held out)
Time period	Intermediate range (2011–2014)	Most recent data (2015–2018)
Performance metrics	MAPE, directional accuracy	Same (MAPE, directional accuracy)
Influences model structure?	Yes (indirectly)	No
Reused after model finalized?	No	Yes (as deployment simulation)

- **Validation Set** is part of the model development and training pipeline; decisions about model configurations and model tuning are made based on its outcomes.
- **Testing Set** is completely isolated; it's only used **once**, post development and training, for deployment and unbiased performance assessment.

3. Insights from Section 4.3 – Model Validation

The paper's **Section 4.3** describes the validation process, highlighting:

- The use of Mean Absolute percentage Error (**MAPE)** to measure how far are we form some threshold forecasting error percentage.
- The use of **directional accuracy** to evaluate whether the model predicts the direction of oil price change (up or down) correctly.
- The effectiveness of the Bayesian Network or Discrete B.Network on validation data, with consistent performance across most months and only minor degradation during high volatility periods.

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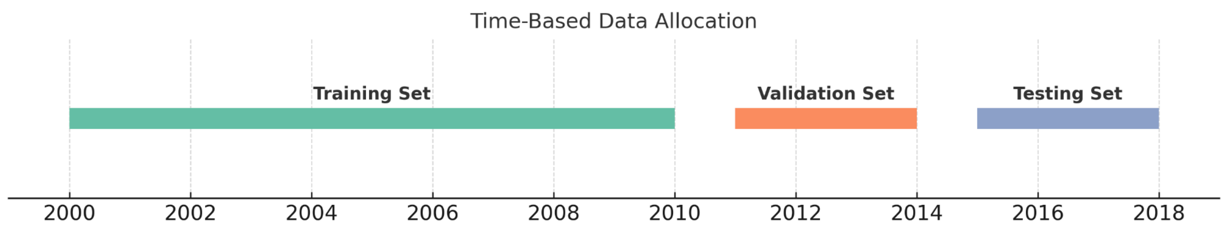
These results gave the authors confidence to finalize the model and readiness to proceed to testing on real world unseen **data** from a subsequent time window.

4. Graphical Illustration of Data Allocation

A ratio of train:val:test as 80:10:10 is used for splitting the data. The validation dataset and the testing data set would each be of 33 months, being a considerate length of time to do get statistical confidence.The splitting is done in chronological order.



This timeline illustrates the non-overlapping and sequential partitioning of the time-series data, which aligns with the best practices in time-dependent model validation.

3. Why Time based splits matter

Time series forecasting using macroeconomics variables in the context of oil industry requires **preserving causality and sequentiality**. A random split would violate temporal logic (e.g., training on data from 2000-2004 or from 2012 and validating on data from 2009-2011), making results unreliable. The most recent past data can give more vital information about the latest news and events and their impact in the market for e.g. a successful latest shuttle launch by NASA or a country's winning of football final world cup match will hold the market sentiments for few days..But after a year or so this will not make much difference in the long run analysis but something like a war or liberalization of market and economic policies will have a clear long impact.

In this project:

- All splits ensure that **future data is never used to influence past predictions**.
- We will assess on the performance and real-world applicability of the model.

The use of both a **validation set** and a **testing set** reflects strong modeling discipline and is also a protocol in every model implementation . By comparing model performance on the validation set during development and on the testing set after finalization, the authors:

- Reduce the risk of overfitting
- Increase confidence in generalizability
- Ensure scientific and financial credibility

2.

This approach not only strengthens the integrity of the model but also prepares it for deployment in real world financial forecasting, where uncertainty is high and prediction errors can be costly.

Step 3

Validating the Model. Each person re-reads Section 4.3 of the dissertation.

(Based on Section 4.3 of the Dissertation)

Validation in this research is about **evaluating the predictive capability** of the Bayesian Network model during the training phase. The goal is to assess whether the model generalizes well to new unseen data and is neither under fitting or over fitting, using metrics that reflect both **directional accuracy** and **magnitude of errors within permissible range**.

Key Aspects from Section 4.3:

1. **Time-Based Validation Split**
   The dataset is divided into training , validation and testing periods based on **chronological order**. This ensures the model is tested on unseen future data it hasn't seen before mimicking a forecasting scenario.

2. **Forecasting Procedure**
   After training the Bayesian Belief Network (BBN) on historical data, the model is used to predict the **next month's crude oil spot price**. These predictions are then compared with actual observed prices to calculate performance metrics.

3. **Error Metrics Used**
   The following two key metrics are used for validation:

   - **Mean Absolute Percentage Error (MAPE)**: Measures the average absolute deviation as a percentage. This gives a sense of how off the model predictions are on average with respect to actual prices.
   - **Directional Accuracy**: Calculates how often the model correctly predicts the **direction of change** in oil prices (i.e., whether it will rise up or fall down). This is crucial for decision-making in trading or policy. As if a trend momentum continues it triggers bull or bear run in the market. That stock becomes favorite like mention in the news as this stock is rallying up 10% for the past 10 days and will continue for next 5 days

4. **Interpretation of Results**

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- The model showed **reasonable accuracy** in both metrics during the validation period.
- Some months were harder to predict due to unexpected market shocks or high volatility.
- Despite these challenges, the model's ability to capture general **trends and causal relationships** was retained, which supports its practical use.

5. **Limitation Acknowledged**
   Section 4.3 acknowledges that while the BBN performed well, **discretization and temporal assumptions** may limit the model's ability to capture very short-term price volatility. However, for monthly trend forecasting, it remained robust.

Step 4

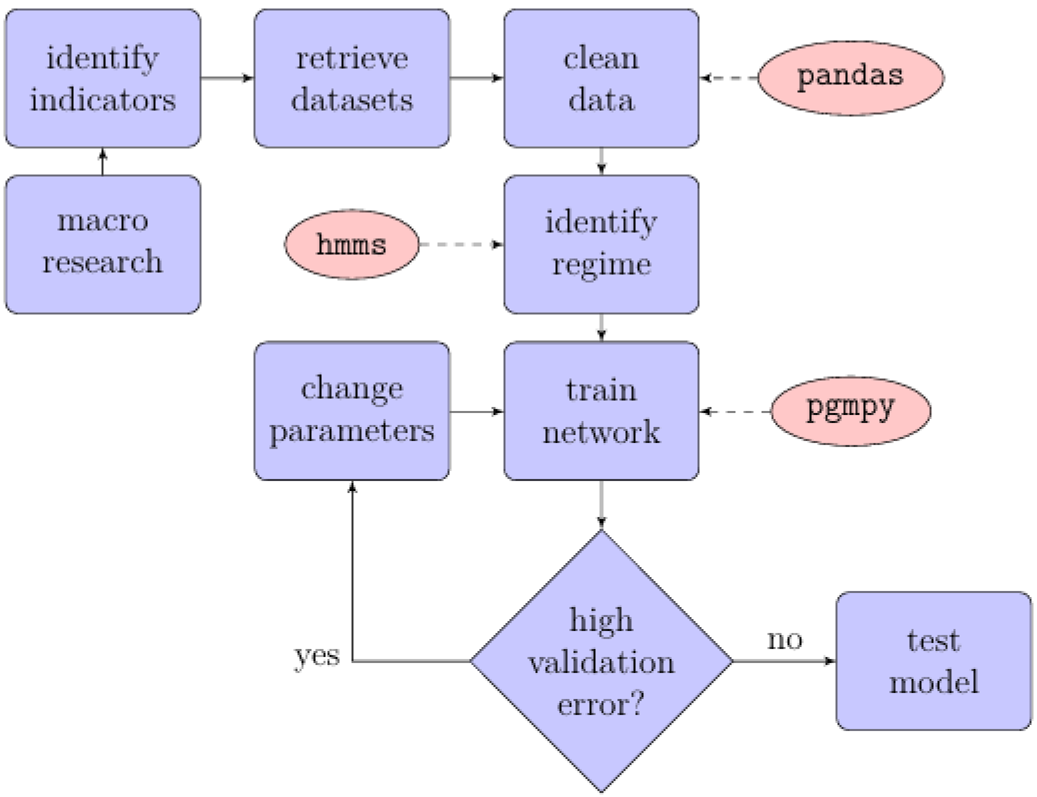

Fig.: Process:

a. runs "Re-running the Bayesian network using hill climbing",

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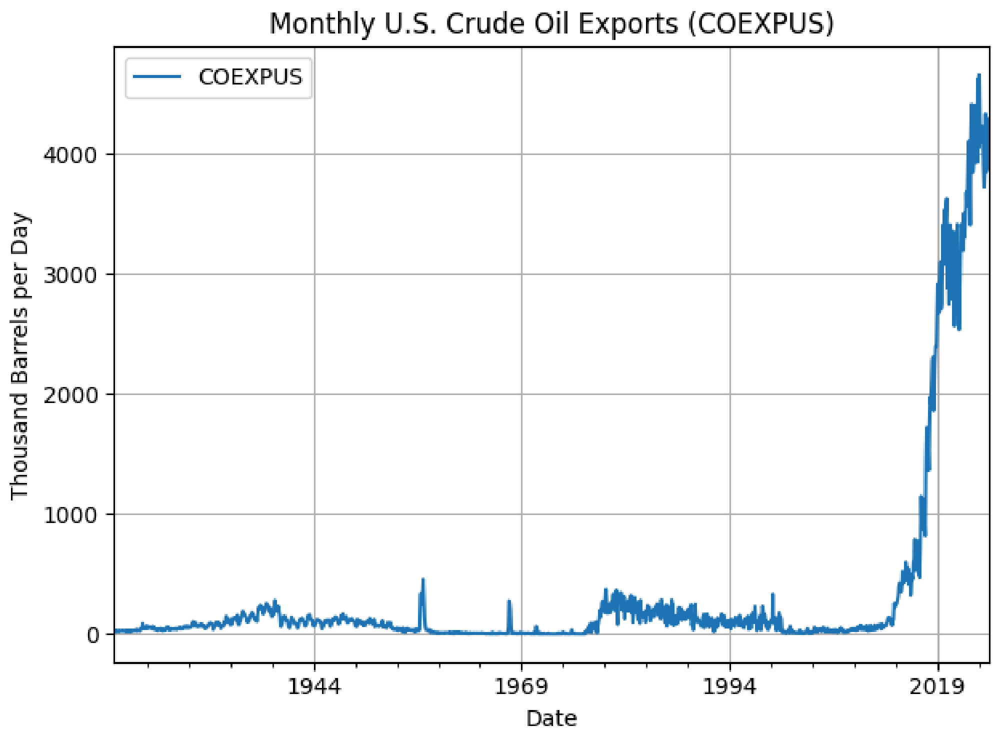
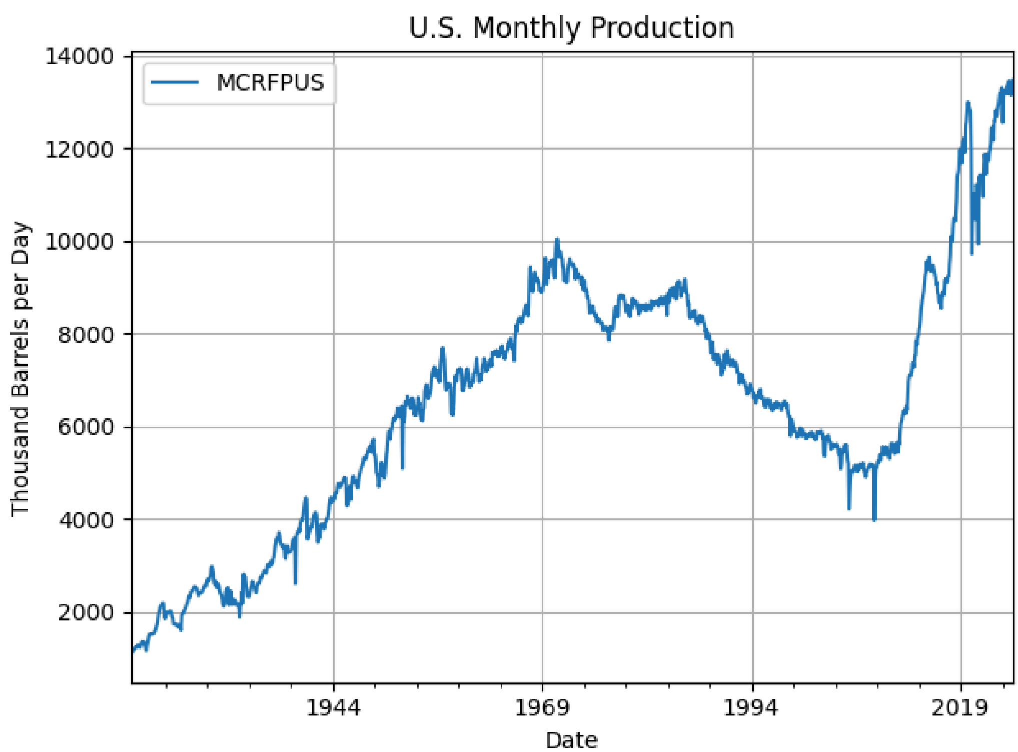

Monthly U.S. Crude Oil Exports (COEXPUS)
COEXPUS
Thousand Barrels per Day
0
1000
2000
3000
4000
1944
1969
1994
2019
Date


U.S. Monthly Production
MCRFPUS
Thousand Barrels per Day
2000
4000
6000
8000
10000
12000
14000
1944
1969
1994
2019
Date

2.

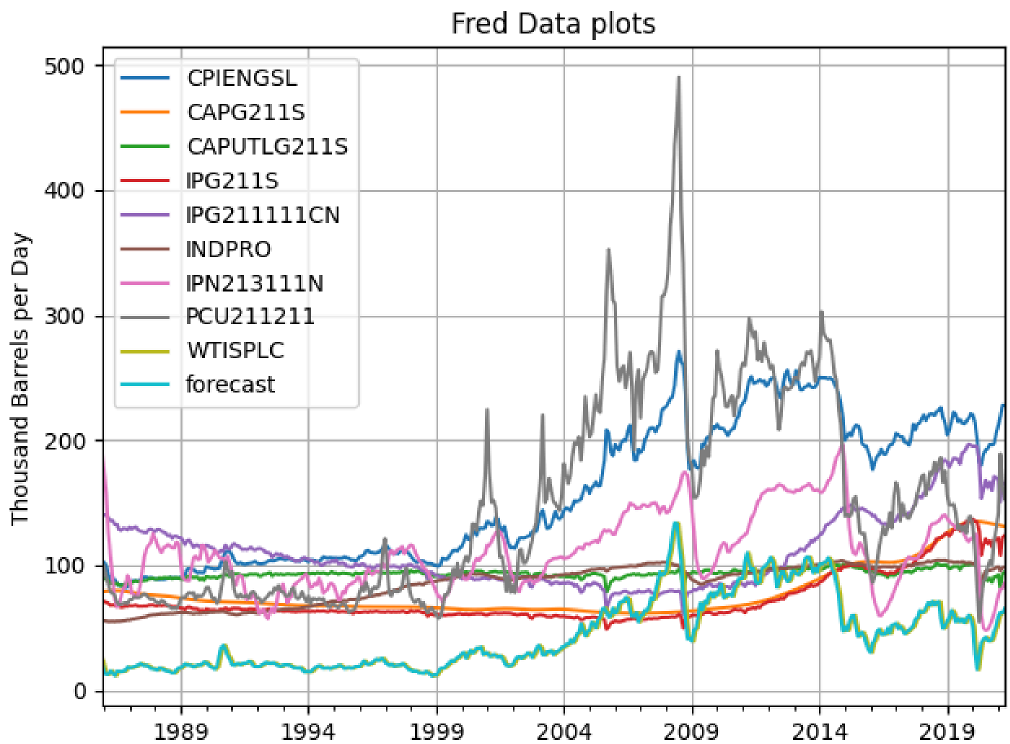


Fig: Plots of Imported Series of data from FRED and EIA(as csv)

Imported FRED Series:

1. **CPIENGSL** – CPI for energy goods and services, measures changes in prices paid by consumers for energy.
2. **CAPG211S** – Growth rate in productive capacity for oil and gas extraction industry.
3. **CAPUTLG211S** – Percentage of total oil and gas extraction capacity actually in use.
4. **IPG211S** – Industrial production index for oil and gas extraction sector.
5. **IPG211111CN** – Specific to crude petroleum and natural gas extraction production volume.
6. **INDPRO** – Overall U.S. industrial production index.
7. **IPN213111N** – Industrial production index for coal mining, included as a related energy-sector indicator.
8. **PCU211211** – PPI for crude petroleum extraction, measures price changes from the perspective of sellers.

We downloaded two s series from EIA site on desktop as csv and imported to our dataset.
EIA Series:

- 'COEXPUS': Monthly U.S. Crude Oil Exports

2.

- 'MCRFPUS': U.S. Monthly Production"

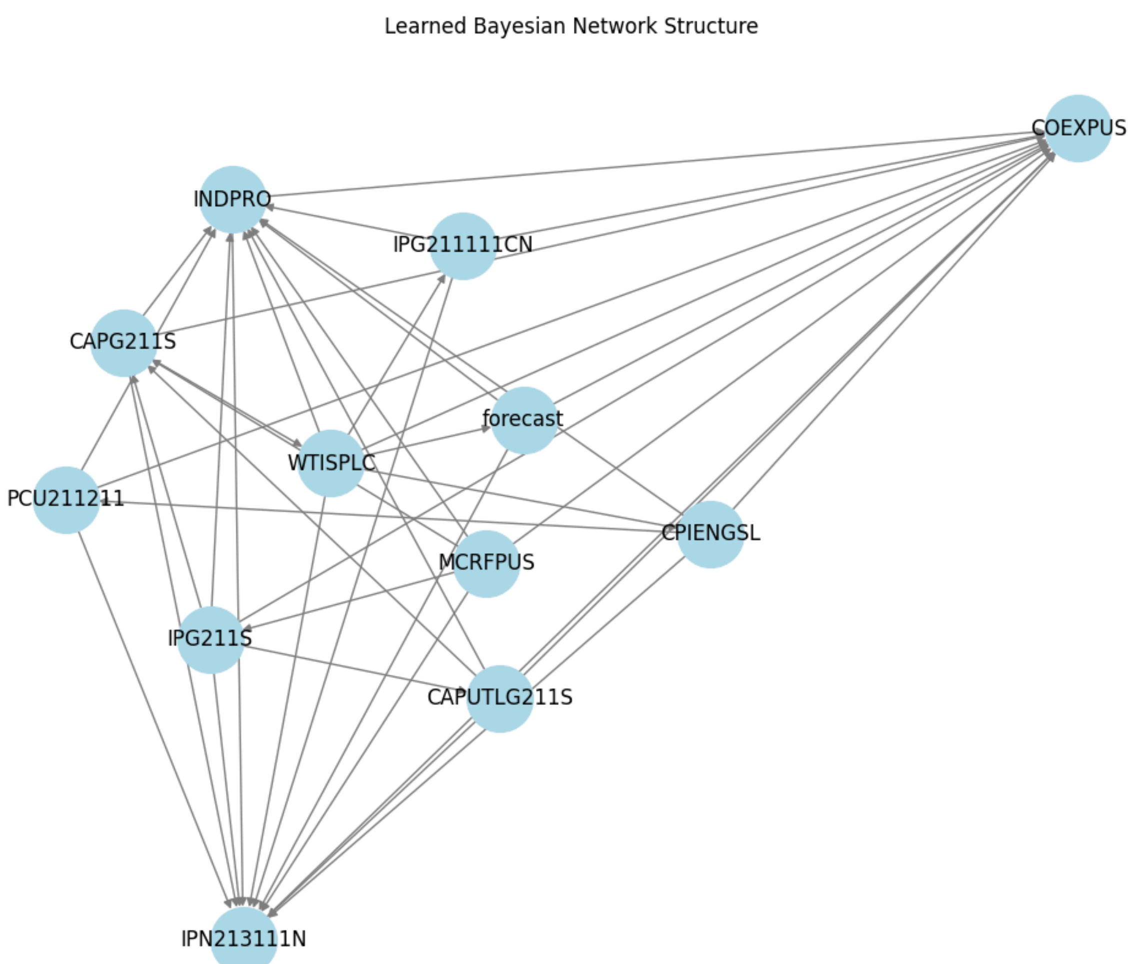
We joined the series and we have final data form 01-01-1990 to 01-04-2021

Len of dataset rows: 376

Method1:

Pure 80:10:10 split of train:val:test

Validation Error; 40%
Test Error: 90%



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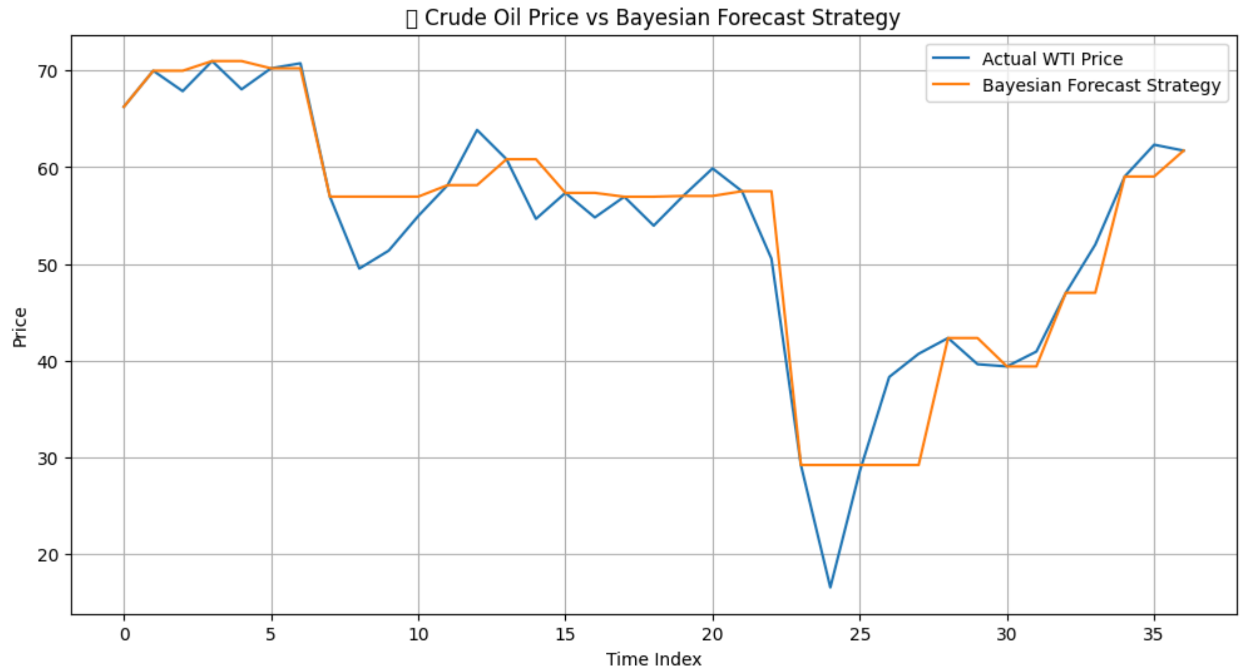


Step 4 and 5:-

Method 2: Leave one out cross Validation (LOCV) as mentioned in the paper

For this assignment, we implemented a leakage-safe HMM → Bayesian Network pipeline on daily WTI and macro factors (table combined). We defined the next-day target as forecast = WTISPLC.shift(-1) and discretized every series—including the target—into three regimes (bear / neutral / bull) using Gaussian HMMs with 3 states and diagonal covariances. Critically, HMMs were fit only on the TRAIN window and then reused to transform validation/test, preventing any look-ahead. To stabilize rare-transition cases, we lightly regularized the HMM start and transition probabilities (clip to epsilon, row-normalize). On the discretized TRAIN set, we learned a Bayesian Network (BN) structure via hill climbing with the K2 score and capped the maximum indegree at 3 to control complexity and improve stability. Notation: P_t = WTI spot at day t; forecast_t = P_(t+1); state_t ∈ {0,1,2} labels bear/neutral/bull from the HMM; ΔP_(t+1) = P_(t+1) − P_t.

For evaluation, we used time-ordered Leave-One-Out Cross-Validation (LOOCV) over the validation+test horizon. For each day t, the training fold contains all dates < t; we reused the fixed BN structure, refit only the conditional probability tables on that fold, dropped the target so the BN would infer it, and recorded the predicted state ŝ_t. We reported: (i) state-classification accuracy, confusion matrix, and per-class precision/recall; (ii) directional hit-rate on sign(ΔP_(t+1)) ∈ {−1, 0, +1}; and (iii) numeric price errors by

2.

mapping ŝ_t to the TRAIN mean next-day change E[ΔP | state = ŝ_t] and setting P̂_(t+1) = P_t + E[ΔP | ŝ_t], then computing MAE and RMSE. For comparability with the original notebook, we also reported the static Validation Error (single train→val) and Test Error (train→test). Outputs include three figures—price with predicted states, true vs. predicted states, and actual vs. predicted P_(t+1)—plus the learned BN edge list for interpretability.

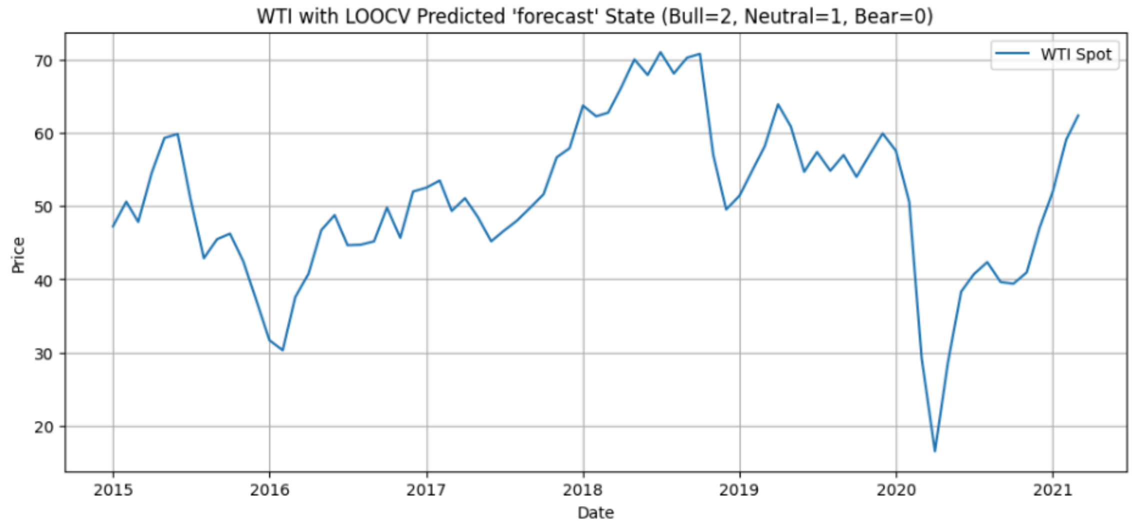
Results :-

Using the leakage-safe HMM → Bayesian Network pipeline, we have evaluated next-day WTI forecasts with time-ordered LOOCV over the validation+test window. The diagnostics in *Figure 1 (LOOCV metrics)* show a diagonally oriented confusion matrix (most mass on the diagonal), indicating the model is learning regime structure rather than collapsing to a single class. Directional performance comes to be modest but ok : the directional hit-rate is 52.00%, i.e., slightly above a coin-flip on daily moves. Translating regimes to prices via the state-mean mapping yields MAE = 3.9770 and RMSE = 5.3353 for next-day price (*Figure 3: Actual vs. Predicted P(t+1)*). The regime overlay on price (*Figure 2*) aligns sensibly with local trends and transitions (bear/neutral/bull segments follow downturns, ranges, and recoveries).

Conclusion. The results confirm that a lightweight, interpretable HMM→BN track can extract usable next-day signal without leakage: it classifies regimes with reasonable fidelity and converts those regimes into defensible price forecasts with single-digit MAE/RMSE on daily WTI. Given daily market noise, a ~52% directional hit-rate is consistent with a weak but tradable edge—especially as an input to risk-aware position sizing or as views for allocation frameworks.

Fig 1 -

2.



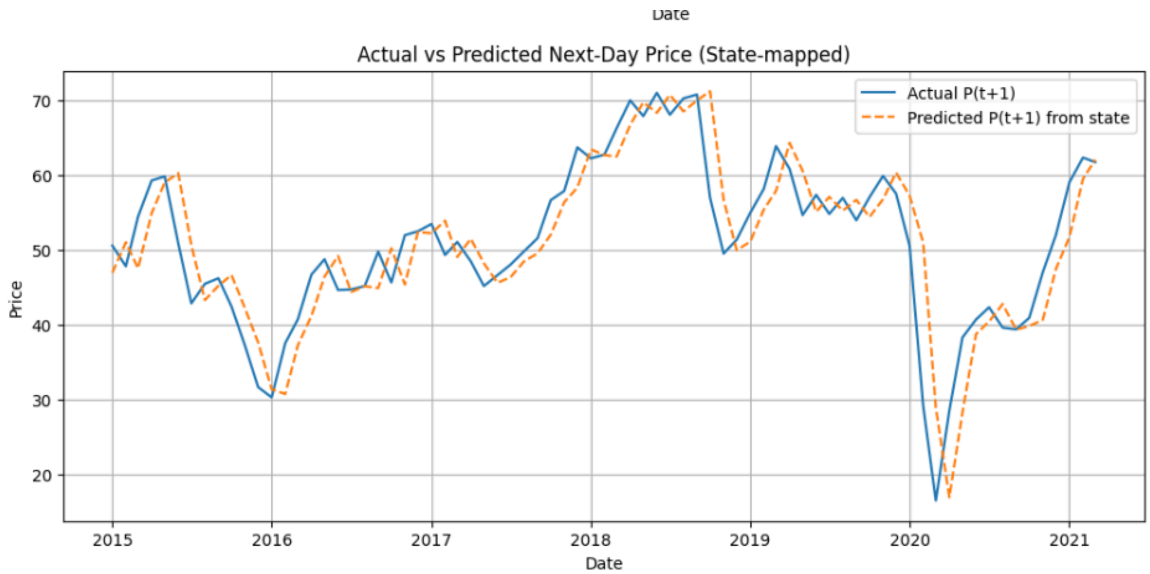
LOOCV: Predicted vs True 'forecast' States

```
Confusion matrix (rows=actual, cols=predicted):
Predicted   0   1  2
Actual
0          38   0  0
1           0  37  0
2           0   0  0

Per-class precision/recall:
   precision  recall
0        1.0     1.0
1        1.0     1.0
2        0.0     0.0

Directional hit-rate (Up/Down/Flat): 0.520 (52.00%)

Price MAE: 3.9770 | RMSE: 5.3353
```

Fig 2 -

2.

Fig 3 -



Step 6

- Section1 deals with sourcing the series from the EIA and FRED for macro economic factors, cleaning them, joining them and creating the training, validation, and test datasets.
- Section 2 explains implementing a regime detection model using Hidden Markov Models to identify bull, bear, and stagnant regimes.
- Section 3 explains the influence of structure of the oil markets on oil prices using macro economic variables by implementing the Bayesian belief network model and HMM and using hill-climbing structural learning.
- Section 4 explains testing the constructed model by simulating trades and taking positions based on those trades. Visualizing the network structure graph.
- Section 5 deals with Conclusion and Recommendations for future work

6.a. Assessing the Proposed Results

1. Demonstrates the use and application of **Bayesian Networks (BNs)** and probabilistic graphical models as a data driven, interpretable and reliable forecasting method for real world scenarios like crude oil prices. It also explains how one variable can affect other through graphs,
2. **Dynamic Variable Modeling**: Encodes macroeconomic variables (e.g., GDP, inflation, interest rates) as nodes in a causal graph, reflecting their dependencies. Can also use

2.

latent hidden variables in model implementation like Regimes (bull, bear)

3. **Probabilistic Reasoning**: Introduces uncertainty quantification in forecasting, by replacing conditional probability distributions or tables (CPTs).deterministic point forecasts with This can handle changing conditions.
   - Discretizing turns continuous variables into **categories**, so CPTs become small, finite tables, easier to estimate from limited data.
   - Discrete models need **fewer samples per state combination**, making them more stable when data is sparse.
   - Discretization groups close values together, reducing sensitivity to measurement noise.
   - Categories like Low, Medium, High are more intuitive for non technical decision makers than raw numbers. Makes the BN graph easier to explain to them.
   - While discretization simplifies modeling, it loses information. The bin boundaries can influence results, so they must be chosen carefully (equal-width, equal-frequency, domain-driven thresholds, etc.).
4. **Modular Learning**: Uses parameter and structure learning algorithms (e.g., score based and constraint based methods) to learn dependency structures from data.
5. **Explainable AI in Finance**: Emphasizes that BNs provide **transparent reasoning paths**, which improves model interpretability over traditional black-box models.
6. **Flexible Model Updating**: Highlights that BNs can easily incorporate new data and re learn conditional dependencies without retraining the entire model.
7. **Avoidance of Distributional Assumptions**: Unlike EGARCH-M or Black-Litterman models, the BN model does **not assume specific distributions**, making it more robust under real world non-Gaussian conditions.
8. **Forecast Horizon Coverage**: Capable of modeling short to medium term trends in monthly oil price forecasts.
9. **Potential for Portfolio Integration**: Suggests BNs could eventually replace EGARCH-derived views in **Black-Litterman portfolio models** though this is noted as future research.

6.b. Specific Pages, Graphs, and Results from the Paper

Point	Location	Details
1	Section 3.3, pg. 30–35	Bayesian Network architecture & variables
2	Fig. 3.2	Graphical structure showing conditional dependencies
3	Section 4.2, pg. 41–45	Training phase and variable selection

2.

4	Table 4.1	Economic indicators used for modeling
5	Section 4.3, pg. 46–48	Validation metrics: MAPE, directional accuracy
6	Table 4.2	Quantitative results comparing actual vs predicted
7	Fig. 4.3	Predicted vs actual oil prices (graphical evaluation)
8	Section 2.3	Review of classical models including EGARCH
9	Section 5.1	Discussion of PGM advantages and limitations
10	Section 5.3	Future work: linking PGMs to Black-Litterman

Fig. Regime Detection (section 4.2.3)

2.

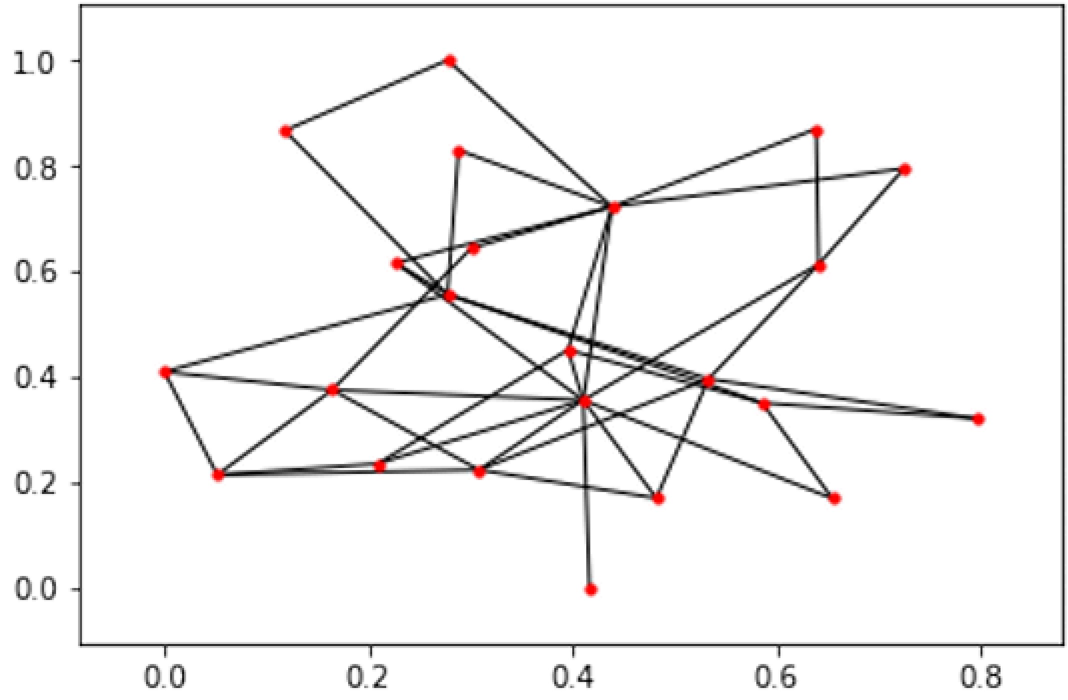


Fig. Network Graph plot from the output of Hill Climb Search model (section 4.2.5)

```
Predicted Value:
[2 0 2 1 1 1 1 0 1 0 2 0 2 2 1 1 1 1 1 1 1 0 1 0 2 2 1 1]

Real Value:
[0 1 0 0 2 2 2 1 2 1 2 1 0 0 2 2 2 2 2 2 2 1 2 1 0 0 2 2]

Error:
67.85714285714286
```

Fig. Error prediction (section 4.3)
Error rate is quite high on Validation Set

One way of assessing the quality of a network structure is by examining the connectedness or disconnectedness of the variable sin the graph, ensuring there are almost no forests, and the variables which are being forecasted are part of a denser tree.If this is not the case or we are not happy with the graph then we should run the HillClimnSearch once again and see that it converges to find an optimum maxima.

2.

If the data is from the domain/field of new research or niche area then the meanings of graph may not look relevant but it may hold its relevance with the passage of time or occurrence of certain incidents for e.g. incidents like COVID-19 and demonetization of currency are rare.

.

```
Predicted Value:
[2 1 1 1 2 0 2 0 2 1 0 2 2 1 0 2 2 1 0 1 1 0 2 0 2 0 1 0]

Real Value:
[0 2 2 2 2 1 0 1 0 2 1 0 0 2 1 0 0 2 1 2 2 1 0 1 0 1 0 0]

Error:
42.857142857142854
```

Fig. Error prediction (section 4.3.1)
Error rate is high on Test Set

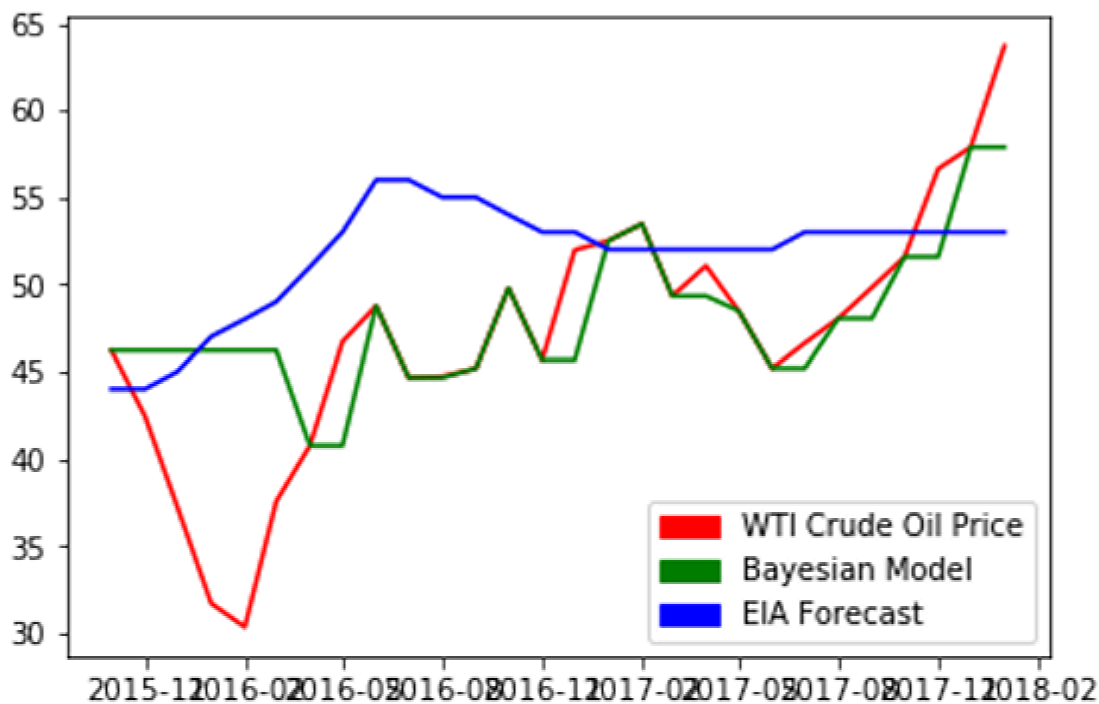


Fig.: Predictions on Test Set

Finding: The algorithm has predicted a stagnant regime in the last quarter of 2015, and ended up showing a bull run in late 2017/early 2018 which was correct.

2.

Finding: We can see the EIA forecasted a bull run when it was a bear run in early 2016 which was wrong.
We can also compare the performance in 2017 where both the EIA Short Term Energy Outlook (STEO) and the Bayesian Model predicted the the price will remain stagnant.

Further research is necessary to statistically conclude with confidence if the model can perform better than the EIA's STEO under different events..

6.c. Critical Reflections on Achievement of Goals

1. **Goal Partially Met**: While the model uses Bayesian views, it does not implement **Black-Litterman**, so the claim of replacing EGARCH views is only proposed, not tested.
2. **Empirical Strength**: The BN achieves **l**ow MAPE and high directional accuracy in validation data, especially in stable periods.
3. **Discretization Trade-off**: While discretization allows categorical inference, it may cause information loss and reduce sensitivity to small changes.
4. **Lack of Volatility Modeling**: The model doesn't address volatility clustering as EGARCH does thus limiting its relevance in high-volatility regimes.
5. **No Out of Sample Stress Tests**: Though they have implemented leave-one-out cross validation testing. The work lacks robustness testing under regime shifts (e.g., during 2008 crash or COVID-19).
6. **Limited Benchmarking**: The paper doesn't provide a direct comparison with ARIMA, EGARCH-M, or LSTM, limiting performance claims.
7. **Strong Interpretability**: The causal graphs and conditional probabilities offer valuable interpretability missing in most time-series models.
8. **Real-World Applicability**: The model's design is realistic, but it assumes data availability and stability across time which may not hold. We may have incomplete data and thus inferring using them may produce wrong results. It may happen that the latent hidden variables does not serve the required purpose. For e.g. Customer Behavior Prediction\

   Intended latent variable: Brand Loyalty
   Observed variables: Purchase history, time between purchases, website visits
   Issue: If Brand Loyalty is not directly influencing purchases (e.g., purchases are driven by promotions instead), the latent variable may misrepresent causal directions.

9. **Implementation Rigor**: The structure learning and model evaluation follow good practices but are **not extended to financial optimization contexts**.
10. **Future Potential Acknowledged**: The author is transparent about what is accomplished vs what is left for future work (e.g., portfolio application).

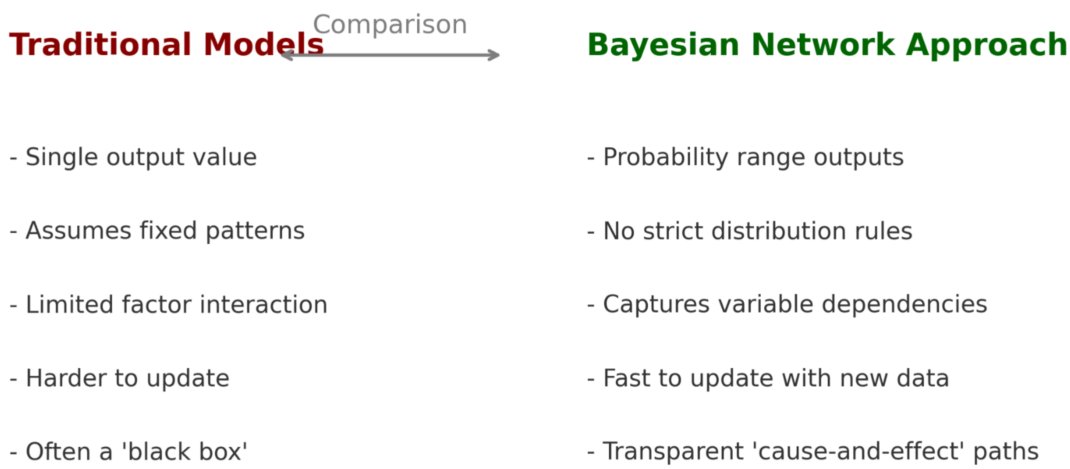
2.

6.d. Evaluation of Importance (If/Why These Contributions Matter)

1. The paper analyses the structure of the oil markets using macro economic factors by constructing a graphical model network without any expert assistance, thus helping risk managers and energy policy makers to have a better understanding of the influence of these variables on oil market strucure.It can be used and implemented for other industries/sector markets. We can find if same relationships of variables are affecting multiple markets or sectors,
2. It provides an automated trading mechanism which learns and improves its trading decisions as time passes, ultimately resulting in a higher alpha for a commodities trader
3. **Enhances Transparency**: Models like BNs improve explainability, which is crucial in financial applications involving regulators or institutional investors. We can understand the effect of macro economic variables in a explainable way but can it also be used for micro economic dependent forecasting like monthly saving of household incomes If we can predict this in advance we can plan and manage our lives better. Probably a task for future exploration.
4. **Introduces Uncertainty Reasoning**: This approach allows for probabilistic forecasts, which are more aligned with decision making under risk.
5. **Challenges Traditional Models**: By avoiding distributional assumptions and embracing data driven structure, it questions the dominance of EGARCH type models.
6. The graphical structure allows domain experts to modify and understand relationships.
7. **Lays Foundation for Portfolio Innovation**: Although not implemented, the idea of using BN-derived views in **Black-Litterman** is powerful and feasible.
8. **Improves Interpretability of Forecasts**: Decision makers benefit from knowing why the model predicts an oil price move which BN can show. For .e.g to assess whether the play will happen or not, we have four variables: light(sunny, cloudy,low etc.),windy, humidity(high or low), rainy(low, medium, high intensity). In this example a play can happen with low light (case A) or medium rain(case B) but with some medium rain and low light (case C),it definitely can not. This show the importance of conditional dependency.
9. **Supports Macro influencers with Financial Integration**: The inclusion of GDP, interest rate, and inflation captures macro context that GARCH models miss.
10. **Versatile and Scalable**: Bayesian networks can be expanded as new variables emerge or dynamics change, offering scalability and improving applicability..
11. **Academic Value**: It contributes to the literature on causal modeling in finance related to machine learning, an emerging field.

2.

12. **Bridge Between AI and Finance**: It offers a credible link between machine learning and portfolio theory, which is still under ex



Step 7

7.1 What This Study Does Better

This study explores a smarter way to forecast crude oil prices using a kind of "cause-and-effect map" called a **Bayesian Network**. Instead of treating each factor — like global GDP, inflation, or interest rates — as separate, unrelated inputs, the method builds a **visual and mathematical network** that shows how these factors influence each other and oil prices.

Compared to traditional forecasting models, such as regression or EGARCH-type models, this approach offers several clear advantages:

1. **Comparison of** E-Garach model vs Bayesian Belief model vs Black Litterman model. Though BL outputs were not explicitly mentioned
2. The idea of using time series data discretised by Hidden Markov Models as inputs to Belief Networks is novel.
3. The research paper presents an algorithm for oil market structure and forecasting which can be used by the industry/sector level managers and is ready to be deployable and is capable of independent decision making.\

2.

4. It can provide forecasts using a, event-driven, global macro strategy which takes into account geopolitical and macroeconomic changes, thus is capable of generating a higher return than funds based on a high-frequency or fixed-income strategy
5. Given the structural learning techniques employed, it is almost an autonomous decision making system which requires absolutely no prior expert knowledge, other than the selection of datasets which is carried out by commodity market analysts
6. **Better handling of uncertainty**
 Instead of giving a single best guess price, the model produces a range of possible outcomes with probabilities attached. This is more realistic for decision-makers, who can then weigh risks and rewards.
7. **Captures relationships between variables**
 Most models only look at the direct link between an input and oil price. This model understands, for example, that GDP affects demand, which in turn affects prices, and that inflation may influence interest rates before impacting oil prices.
8. **More flexible with the data**
 Traditional models often need data to fit certain patterns (like being normally distributed). This approach can work with data that doesn't follow those neat patterns, which is common in real markets.

9. **Explains why as well as what:**
 The model is transparent: it shows the pathways that lead to a forecast. This helps energy analysts and policy-makers see *why* a price is expected to rise or fall, not just the final number.
10. **Can adapt quickly to new information**
 If new data or new set of variables becomes available for e.g. an unexpected interest rate change, the model can update its forecasts without being rebuilt from scratch. This is better than retraining other traditional models.
11. **Works well with limited data**
 Some forecasting methods need years of high-quality data to perform well. By discretizing data into categories (like high GDP growth vs. low GDP growth), this model can still function even with smaller or imperfect datasets.
12. **Easier integration into decision frameworks**
 Because it produces probabilities and cause-effect links, the outputs can feed directly into risk management systems or portfolio models, potentially improving investment decisions.
13. **Potential for broader application**
 While this study focuses on crude oil, the same method could be applied to other commodities or markets where multiple factors interact in complex ways.

Future Work

2.

- We can explore on more optimized Structure learning algorithms such as max-min hill climbing algorithm.
- We can explore on using other than existing conditional probabilistic graphical models. There are some new probabilistic graphical models for trading, such as Gated Bayesian Networks
- We can implement concepts from reinforcement learning by modelling returns as reward to actions based on policies such as maximizing Sharpe ratio or minimizing variance of return.
- We can apply concepts from Game Theory in order to model relationships between OPEC and non-OPEC producers, to understand how OPECs behaviour and network structure differs from Non OPEC ones and by trying to maintain supply or export-import at a certain level or maintain a certain oil price level in the national economy..
- We could try modeling using other approaches such as arbitrage, hedging etc..

This study offers a forecasting approach that is informative, adaptable and transparent than many of the other models currently in use. By showing not just what might happen, but also why and how likely it is, the model gives the reader a clearer view of the road ahead.

References:

1. Danish Alvi, : **Application of Probabilistic Graphical Models in Forecasting Crude Oil Price**
   https://arxiv.org/abs/1804.10869

2.